Tutorial on 10 October 2025

(7) $\mathcal{L} = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$

(1) Write down the Hamiltonian

(2) Write Hamiltonian EOM

(3) Solve the EOM

(4) Make plot of p as function of x or phase space diagram

Sol^n $p = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}$ $\qquad \dot{x} = \frac{p}{m}$

$$\mathcal{H} = \frac{p^2}{2m} + \frac{k x^2}{2}$$

HEOM $\Rightarrow$ $\dot{p} = -\frac{\partial \mathcal{H}}{\partial x}$

$$\dot{x} = \frac{\partial \mathcal{H}}{\partial p}$$

$\dot{x} = \frac{p}{m}$ and $\dot{p} = -kx$

or $\ddot{x} + \frac{k}{m} x = 0$

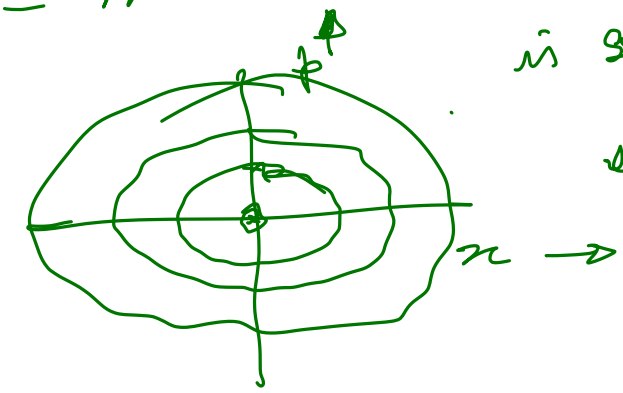
$$\omega = \sqrt{\frac{k}{m}}$$

$$x = A \cos \omega t$$

$$\dot{x} = -A\omega \sin \omega t$$

$$p = -A\frac{\omega}{m} \sin \omega t$$

$$x^2 + \frac{m^2}{\omega^2} p^2 = A^2$$



is set of ellipse

because H is constant

$$H = \frac{p^2}{2m} + k x^2$$

is orbits of ellipse

it is not needed to solve.

Problem 2

⊛ Consider the Lagrangian of the form

$$\mathcal{L} = m\dot{x}\dot{y} + \frac{\gamma}{2}\cdot(x\dot{y} - \dot{x}y) - kxy$$

is called Bateman Lagrangian

⊛ find Lagrange EOM (second order ODE)

⊛ Write down the Hamiltonian

Sol^n

$$\mathcal{L} = m\dot{x}\dot{y} + \frac{\gamma}{2}(x\dot{y} - \dot{x}y) - kxy$$

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{y} - \frac{\gamma}{2}y \qquad p_y = m\dot{x} + \frac{\gamma}{2}x$$

EOM for x $\Rightarrow$ $m\ddot{y} - \frac{\gamma}{2}\dot{y} - \frac{\gamma}{2}\dot{y} + ky = 0$

$$m\ddot{y} - \gamma\dot{y} + ky = 0 \;!$$

EOM for y $\Rightarrow$ $m\ddot{x} + \frac{\gamma}{2}\dot{x} + \frac{\gamma}{2}\dot{x} + kx = 0$

$$\frac{1}{m}\left(p_x + \frac{\gamma}{2} y\right) = \dot{y} \qquad \frac{1}{m}\left(p_y - \frac{\gamma}{2} x\right) = \dot{x}$$

$$\mathcal{H} = p_x \dot{x} + p_y \dot{y} - \mathcal{L}$$

$$\mathcal{H} = p_x \dot{x} + p_y \dot{y} - m \dot{x} \dot{y} - \frac{\gamma}{2} \cdot (x \dot{y} - \dot{x} y) + k x y$$

$$= \frac{1}{m} p_x \left(p_y - \frac{\gamma}{2} x\right) + \frac{1}{m} p_y \left(p_x + \frac{\gamma}{2} y\right)$$

$$- \frac{1}{m}\left(p_y - \frac{\gamma}{2} x\right)\left(p_x + \frac{\gamma}{2} y\right)$$

$$- \frac{\gamma}{2m} x \left(p_x + \frac{\gamma}{2} y\right) + \frac{\gamma}{2m} y \left(p_y - \frac{\gamma}{2} x\right) + k x y$$

$$= \frac{1}{m} p_x p_y - \cancel{\frac{\gamma x p_x}{2m}} + \cancel{\frac{1}{m} y p_x} + \cancel{\frac{\gamma y}{2m} p_y}$$

$$- \cancel{\frac{1}{m} p_x p_y} - \cancel{\frac{\gamma y p_y}{2m}} + \cancel{\frac{\gamma x p_x}{2m}} + \frac{\gamma}{4} x y + k x y$$

$$H = \frac{1}{2} p_x p_y + \frac{\gamma}{4} x y + k x y$$

Problem 3

Solve the differential Equation

① $$\frac{d^2y}{dt^2} + \omega^2 y = k$$

k and ω^2 are constants

② $$\frac{d^2y}{dx^2} + 2\gamma \frac{dy}{dx} + \omega^2 y = F \sin \Omega x$$

Solⁿ The homogeneous part of both ode is

$$\frac{d^2y}{dx^2} + \omega^2 y = 0 \text{ is}$$

$$y_g = A \sin \omega x + B \cos \omega x$$

general solⁿ is $y = y_g + y_p$

in this case y_p is particular solⁿ

$y_p = k$ it can be substituted and obtained

$$y_p = k$$

Now Sol^n is part ① is

$$y = A\sin\omega x + B\cos\omega x + k$$

Part 2

$$\frac{d^2y}{dx^2} + 2\gamma\frac{dy}{dx} + \omega^2 y = F\sin\Omega x$$

The sol^n to the homogenous Eq. can be written from

$$D^2 + 2\gamma D + \omega^2 = 0$$

$$D = -\gamma \pm \sqrt{\gamma^2 - \omega^2}$$

Sol^n is $y = e^{-\gamma x}\cdot(A\sin\omega x + B\cos\omega x)$

Now from inspection the particular sol^n is of the form

$$y_p = A\sin(\Omega x + \delta)$$

and y_p satisfy the Eq^n

$$\frac{d^2y}{dx^2} + 2\nu\frac{dy}{dx} + \omega^2 y = F\sin\Omega x$$

$$-A\Omega^2\sin(\Omega x+\delta) + 2\nu A\Omega\cos(\Omega x+\delta)$$
$$+\ \omega^2 A\sin(\Omega x+\delta) = F\sin\Omega x$$

$$A(\omega^2-\Omega^2)\sin(\Omega x+\delta)$$
$$+\ 2\nu A\Omega\cos[\Omega x+\delta]$$
$$= F\sin\Omega x$$

This we write as

$$\rho\cos\delta'\sin(\Omega x+\delta)$$
$$+\ \rho\sin\delta'\cos(\Omega x+\delta) = F\sin\Omega x$$

$$\rho\cos\delta' = A(\omega^2-\Omega^2)$$

$$\rho\sin\delta' = 2\nu A\Omega$$

$$\rho^2 = A^2(\omega^2 - \Omega^2)^2 + 4A^2\nu^2\Omega^2$$

$$\tan\delta' = \frac{\dot{a}\nu^2\Omega^2}{\omega^2 - \Omega^2}$$

$$\delta = -\delta'$$

$$F^2 = \rho^2$$

is the soln !